Mathematical Foundations of Reinforcement Learning
Homework 3

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Question 1

Consider a shortest-path MRP denoted by $(\mathcal{S}, P, R)$, where n is there initial state and:

- $\mathcal{S} = \{0, 1, 2, \ldots, n\}$ represents the set of states, with state 0 being a reward-free terminal state.
- P is the transition probability matrix, where P_{ij} represents the probability of transitioning from state i to state j.
- R is the reward function, where $R(s)$ represents the immediate reward at state s.

We assume the MRP to be *proper*, meaning that it satisfies the condition:

$$\rho := \max_{s \in \{1,2,\ldots,n\}} \mathbb{P}(S_n \neq 0 \mid S_0 = s) < 1. \tag{1}$$

Given a random trajectory $\omega = (S_0, S_1, S_2, \ldots)$, we define $T(\omega)$ as the hitting time to the terminal state 0, formally expressed as:

$$T(\omega) := \min\{t \in \mathbb{N} : S_t = 0\}. \tag{2}$$

Similarly, we define $C(\omega)$ as the return of the episode, which is the cumulative reward received until the trajectory hits the terminal state, given by:

$$C(\omega) = \sum_{t=0}^{T_e} R(S_t). \tag{3}$$

The objective of this homework is to demonstrate that the variance of the return, $\mathbf{Var}(C(\omega))$, is bounded above by some finite value σ^2, i.e., $\mathbf{Var}(C(\omega)) < \sigma^2$.

a) Consider two discrete random variables $X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ over the countable sample set Ω. Suppose that $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$. Prove that $\mathbf{E}X^2 \leq \mathbf{E}Y^2$.

b) Let $R_{\max}$ be an upper bound on the reward, i.e., $|R(s)| \leq R_{\max}$ for all $s \in \mathcal{S}$. Prove that $|C(\omega)| \leq R_{\max} T(\omega)$ for all $\omega \in \Omega$, where Ω is the set of all trajectories.

c) Prove that $\mathbf{P}(T \geq t) \leq (\rho^{1/n})^t$.

d) Prove that

$$\mathbf{E}T^2 \leq \frac{\rho^{1/n}(\rho^{1/n}+1)}{(1-\rho^{1/n})^3}.$$

Hint: The following series results might be useful

$$\sum_{t=0}^{\infty} t^2 \alpha^t = \frac{\alpha(\alpha+1)}{(1-\alpha)^3},$$

for alpha $\alpha \in [0, 1)$.

e) Conclude that

$$\mathbf{Var}\, C \leq R_{\max}^2 \frac{\rho^{1/n}(\rho^{1/n}+1)}{(1-\rho^{1/n})^3} - V(n)$$

Question 2

Consider a two-state shortest-path MRP $(\mathcal{S}, P, r)$ where $\mathcal{S} = \{0, 1\}$ and 0 is a zero-reward terminal state, i.e., $r(0) = 0$ and $\mathbf{P}(S_{t+1} = 0 | P_t = 0) = 1$. Define $p := \mathbf{P}(S_{t+1} = 0 | P_t = 1)$. We assume that $p < 1$ and the initial state is always 1, i.e., $\mathbf{P}(S_0 = 1) = 1$.

We define the value of the state 1 as

$$v = \mathbf{E}\left[\sum_{t=0}^{\infty} r(S_t) | S_0 = 1\right].$$

In this problem we compare the performance of the first visit and ever visit Monte Carlo for estimating v.

a) Prove that $v = 1/p$.

b) **First Visit MC:** Suppose that we run E episodes $e = 1, 2, \ldots, E$. Let $T(e)$ be the hitting time of the terminal state 0, i.e.

$$T(e) := \min\{t \in \mathbb{N} : S_t = 0\}.$$

Then the return at episode e is

$$C(e) = \sum_{t=0}^{T(e)-1} 1 = T(e).$$

The first visit MC estimation is then

$$\hat{v}_{\mathrm{FV}}(E) = \frac{1}{E}\sum_{e=1}^{E} C(e).$$

Prove that $C(e)$ has mean $1/p$ and variance $(1-p)/p^2$. Conclude that the estimator $\hat{v}_{\mathrm{FV}}(E)$ is an unbiased with the mean squre error

$$\mathtt{MSE}(\hat{v}_{\mathrm{FV}}(E)) = \mathbf{E}(\hat{v}_{\mathrm{FV}}(E) - v)^2 = \frac{1-p}{Ep^2}.$$

c) **Every Visit MC:** Suppose now that we just run one episode with the stooping time T. Show that the every visit MC estimation from the trajectory is

$$\hat{v}_{\mathrm{EV}} = \frac{T+1}{2}$$

and has the mean square error

$$\mathtt{MSE}(\hat{v}_{\mathrm{EV}} = \mathbf{E}(\hat{v}_{\mathrm{EV}}(E) - v)^2 = \frac{1}{2p^2} - \frac{3}{4p} + \frac{1}{4}.$$

d) Do some reasonable computation experiment comparing the two estimators on this 2 state MDP.

Question 3

Consider a discounted MRP with finite state spaces $\mathcal{S}$ and with transition function $P(s'|s)$, reward function $r : \mathcal{S} \to [0, R]$ and discount factor $\gamma \in [0, 1)$. Suppose that for each state $s \in \mathcal{S}$ we can sample a next state, i.e., $S' \sim P(\cdot|s)$. Suppose that Markov chain $(\mathcal{S}, P)$ has a stationary distribution π, i.e., $\pi^T P = P$.

We consider the following synchronous TD-learning algorithm for learning the value function

$$v(s) = \mathbf{E}\left[\sum_{k=0}^{\infty} \gamma^k r(S_k) | S_0 = s\right].$$

Let $V_k \in \mathbb{R}^{|\mathcal{S}|}$ be the estimate of the value function v at iteration k, initialized with $V_0 = \mathbf{0} \in \mathbb{R}^{|\mathcal{S}|}$. For each $k \geq 0$, the update for V_{k+1} is defined componentwise for each state $s \in \mathcal{S}$ by sampling $S'_k(s) \sim P(\cdot \mid s)$ and updating the value of state s as follows:

$$V_{k+1}(s) = V_k(s) + \alpha\left(r(s) + \gamma V_k(S'_k(s)) - V_k(s)\right).$$

This update can be compactly represented in vector form:

$$V_{k+1} = V_k + \alpha\left(r + \gamma U_k V_k - V_k\right),$$

where $R_k = [R_k(s)]_{s\in\mathcal{S}} \in \mathbb{R}^{|\mathcal{S}|}$ represents the vector of rewards for each state s, and $U_k \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}$ is a stochastic matrix where

$$U_k(s,s') = \begin{cases} 1 & \text{if } S_k'(s) = s' \\ 0 & \text{otherwise.} \end{cases}$$

In this problem we prove convergence of synchronous TD-learning. To that end, we consider the diagonal matrix $D = \texttt{Diag}(\pi(1), \pi(2), \dots \pi(|\mathcal{S}|))$ and the weighted inner product $\langle \cdot, \cdot \rangle_D : \mathbb{R}^{|\mathcal{S}|} \times \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}$ defined as

$$\langle x, y \rangle_D = \langle x, Dy \rangle = \sum_{s=1} \pi(s)x(s)y(s).$$

We will consider the weighted norm $||\cdot||_D : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}$ defined by

$$||x||_D = \sqrt{\langle x, x \rangle_D} = \sqrt{\sum_{s=1} \pi(s)x(s)^2}.$$

a) Consider the Bellman operator $T : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ where

$$T(v) = r + \gamma P v.$$

Show that the algorithm can be written as

$$V_{k+1} = V_k + \alpha(T(V_k) - V_k) + \alpha E_k$$

where $E_k = R_k + \gamma U_k V_k - T(V_k)$.

b) Prove that $\mathbf{E}E_k = 0$ and that there exists σ^2 such that

$$\mathbf{E}||E_k||_D^2 \leq \sigma^2 \quad \text{for all} \quad k.$$

Hint: For the second part, one strategy would be to first prove by induction that

$$||V_k||_\infty \leq \frac{R}{1-\gamma} \quad \text{for all} \quad k.$$

c) Prove that $T(v)$ is γ-contractive in the norm $||\cdot||_D$ with the fixed point v. It could be useful to first prove first that $||Pw||_D \leq ||w||_D$ for all $w \in \mathbb{R}^{|\mathcal{S}|}$.

d) Prove the following equation

$$\begin{aligned}\mathbf{E}||V_{k+1} - v||_D^2 =& \mathbf{E}||V_k - v||_D^2 - 2\alpha\mathbf{E}\langle V_k - T(V_k), V_k - v\rangle_D + \alpha^2\mathbf{E}||T(V_k) - V_k||_D^2 \\ &+ \alpha^2\mathbf{E}||E_k||_D^2.\end{aligned}$$

You may use the Polarization identity that states that for all $x, y \in \mathbb{R}^{|\mathcal{S}|}$ we have

$$||x - y||_D^2 = ||x||_D^2 + ||y||_D^2 + 2\langle x, y \rangle_D.$$

e) Prove that following inequalities hold for all $w \in \mathbb{R}^{|\mathcal{S}|}$

$$\langle w - T(w), w - v\rangle_D \geq (1-\gamma)||w-v||_D^2. \tag{4}$$
$$||T(w) - w||_D^2 \leq 4||w-v||_D^2 \tag{5}$$

For the first inequality, the Cauchy–Schwarz might be useful.

f) Use parts c) and d) to establish that

$$\mathbf{E}||V_{k+1} - v||_D^2 \leq \left[1 - 2\alpha(1-\gamma) + 4\alpha^2\right] ||V_k - v||_D^2 + \alpha^2\sigma^2$$

for all k.

g) Prove that if a sequence $z_k \in \mathbb{R}_+$ satisfies the following in inequality

$$z_{k+1} \leq \left[1 - 2\alpha(1-\gamma) + 4\alpha^2\right] z_k + \alpha^2\sigma^2$$

then holds that

$$z_k \leq \left[1 - 2\alpha(1-\gamma) + 4\alpha^2\right]^k z_0 + \frac{\alpha\sigma^2}{2(1-\gamma) - 4\alpha}.$$

Question 4

Consider a discounted MRP with finite state spaces $\mathcal{S}$, transition function $P(s'|s)$, reward function $r : \mathcal{S} \to [0, R]$ and discount factor $\gamma \in [0, 1)$. Suppose that Markov chain $(\mathcal{S}, P)$ has a stationary distribution π, i.e., $\pi^T P = P$.

Our goal is to estimate the value function

$$V(s) = \mathbf{E}\left[\sum_{t=0}^{\infty} \gamma^t r(S_t) \big| S_0 = s\right].$$

We apply the linear function approximation

$$V(s; w) = \phi(s)^{\mathrm{T}} w$$

or in matrix form

$$V_w = \Phi w$$

where

$$\Phi = [\phi(1), \ldots, \phi(S)]^{\mathrm{T}} = \begin{bmatrix} \phi(1)^{\mathrm{T}} \\ \vdots \\ \phi(S)^{\mathrm{T}} \end{bmatrix}.$$

For a given trajectory

$$S_0, S_1, S_2, \ldots$$

consider linear approximation TD-Learning algorihtm:

$$w_{k+1} = w_k + \gamma_k(r(S_k) + \gamma\phi(S_{k+1})^{\mathrm{T}} w_k - \phi(S_k)^{\mathrm{T}} w_k)\phi(S_k), \tag{6}$$

where $\gamma_k > 0$ is a step-size. We now prove that the algorithm converges in the following steps:

a) Show that we can write the algorithm in matrix form as

$$w_{k+1} = w_k - \gamma_k(A_k w_k - b_k)$$

where $A_k = A(S_k, S_{k+1})$ and $b_k = b(S_k, S_{k+1})$ and

$$A(S_k, S_{k+1}) = \phi(S_k)(\phi(S_k) - \gamma\phi(S_{k+1}))^{\mathrm{T}} \quad \text{and} \quad b(S_k, S_{k+1}) = \phi(S_k)r(S_k).$$

b) Consider the stochastic process $X_k = (S_k, S_{k+1})$. Prove that X_k is a Markov chain and has a stationary distribution.

c) Let $X = (S, S')$ be a random variable where $S \sim \mu(\cdot)$ and $S' \sim P(\cdot|S)$. Prove that

$$A := \mathbf{E}A(X) = \Phi^{\mathrm{T}} D(I - \gamma P)\Phi$$
$$b := \mathbf{E}b(X) = \Phi^{\mathrm{T}} Dr,$$

where $D = \mathtt{Diag}(\mu(1), \mu(2), \ldots, \mu(S))$.

d) Suppose that we follow the TD-algorithm with linear function approximation in Equation (6) on a single trajectory. Use Proposition 4.8 [Page 174] in Bertsekas and Tsitsiklis to prove that the algorithm converges to a point with probability 1. Discuss properties of the limit point.